

International Trade Liberalization and Domestic Institutional Reform

Effects of WTO Accession on Chinese Internal Migration Policy

Yuan Tian
University of Nottingham

Presented by Christopher

The Review of Economics and Statistics, May 2024, 106(3): 794–813

Outline

- 1 Motivation and Research Question
- 2 Institutional Background: The Hukou System
- 3 Data: Measuring Migration Regulations
- 4 Trade Shock: China's WTO Accession
- 5 Identification Strategy
- 6 Main Results
- 7 Heterogeneous Effects
- 8 De Facto Effects: Migrant Well-Being
- 9 Trade, Regulation, and Migrant Flows
- 10 Robustness and Extensions
- 11 Discussion and Conclusion

Why Does This Paper Matter?

Central Thesis:

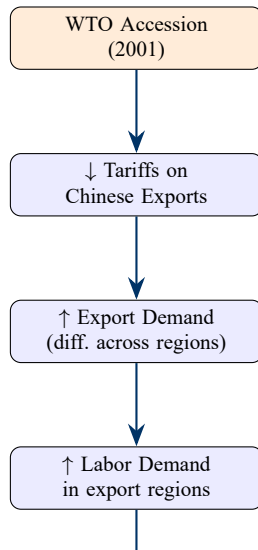
Trade liberalization increases the **effective cost** of maintaining economic institutions that impede factor reallocation.

The Setting:

- ▶ China's WTO accession (Nov. 2001) → large reduction in tariffs on Chinese exports
- ▶ China's Hukou system: a legacy institution regulating internal migration
- ▶ Around 2001: central gov't allowed prefectures to reform Hukou locally

Research Question:

Did export-market liberalization cause Chinese local



Contributions to the Literature

- 1 **Trade → Institutions:** Direct empirical evidence that trade liberalization causes changes in observed regulations
 - ▶ Complements Acemoglu et al. (2005), Trefler & Puga (2014), Khandelwal et al. (2013)
 - ▶ Unlike inferring institutional change from trade patterns, this paper **directly measures** regulatory changes using text data
- 2 **Novel Dataset:** First prefecture-level panel of migration regulations (1978–2016)
 - ▶ 4,273 migrant-related regulations, hand-coded + NLP validation (RF, MLP, CNN)
- 3 **Labor Mobility as Endogenous:** Labor market frictions respond to economic incentives
 - ▶ Extends Tombe & Zhu (2019), Fan (2019), Ma & Tang (2020) who take frictions as given
- 4 **Fiscal Competition:** Prefectures compete to attract migrants by providing amenities
 - ▶ Related to Tiebout (1956), Fajgelbaum et al. (2019)

The Hukou System: China's Internal Migration Barrier

What is Hukou?

- ▶ Internal registration system: each person tied to a **location** and **sector** (agricultural vs. non-agricultural)
- ▶ Living outside Hukou prefecture requires *temporary registration*
- ▶ Even legal migrants face diminished access to:
 - ▶ Medical insurance
 - ▶ Public schools for children
 - ▶ Formal employment

Scale:

- ▶ 2000: 11% of population lived outside Hukou prefecture
- ▶ 2010: 260 million migrants ($\approx 2 \times$ the 2000 figure)
- ▶ In 2016: 22.5% of world population subject to

Timeline of Reform:

Pre-2000: Rigid central control.
Difficult to switch Hukou.

10th 5-Year Plan: “Orderly and
reasonable allocation of labor”

2001: WTO accession. State
Council allows local discretion.

2002–07: 179 of 340 prefectures
enacted 748 new regulations

By 2010: 260M migrants; point-
based systems emerging

Data Collection: A Novel Dataset on Hukou Policies

Source: pkulaw.com — most comprehensive database on Chinese government regulations

Covers 332 of 340 prefectures; keyword search on: *non-Hukou population, migrant worker, temporary residence, Hukou*

Step 1: Identify migrant-related regulations

→ 4,273 regulations (1978–2016)

2,915 prefecture-level; 1,358 province-level

Step 2: Classify by topic:

- 1 Work-related (wages, contracts, training)
- 2 Social insurance (medical, pension, injury)
- 3 Local public goods (schools, vaccination)
- 4 Hukou admin (fees, procedures)
- 5 Family planning
- 6 Miscellaneous

Step 3: Score migrant-friendliness on

$\{-2, -1, 0, 1, 2\}$:

- +2 Complete, executable guideline on specific issue (e.g., mandatory wage deposits)
- +1 Enforcement of existing rules (e.g., timely payment before Chinese New Year)
- 0 Purely administrative/logistic
- 1 Enforcement restricting access
- 2 Guideline depriving migrant rights

Step 4: Cumulate scores to year t :

Validation: Machine Learning Coding

Concern: Subjectivity of manual coding

Solution: Triple-coding (author + 2 RAs) + 3 NLP methods on 686 training regulations

	Random Forest	MLP	CNN w/ Embedding
Method type	Decision tree	Neural net	Neural net
Text features	Words + freq.	Words + freq.	Words + context
Training accuracy	684/686	652/686	624/686
Corr. w/ manual	0.734	0.728	0.712

All NLP codings produce **very similar** empirical results → robust to coding methodology

Key Words from RF Classification:

Top importance: *floating population, migrant worker, temporary residence permit*

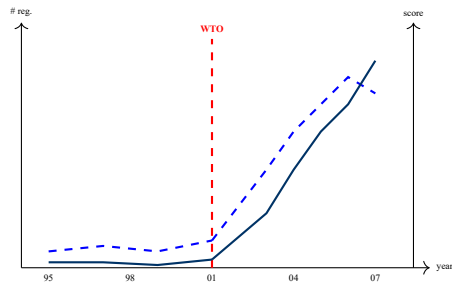
Positive sentiment words: *timeliness, effectiveness, normativity*

Negative sentiment words: *rental housing, public security*

Prefecture-level correlation between manual and RF scores (2001–07 change):
 $\rho = 0.96$

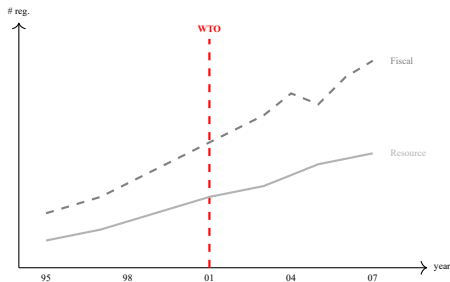
Descriptive Evidence: A Structural Break After 2001

Figure 2(a): Regulation Trends



Pre-2001: ~ 0.07 new regs/prefecture/year; score ≈ 0
Post-2001: Both # and friendliness score **increase sharply**

Figure 2(c): Placebo Regulations



Fiscal and resource regulations show **no structural break** around 2001 $\rightarrow$ rules out mechanical data-coverage explanation

Summary (2002–07):

748 new regulations; 179/340 prefectures; mean score = 1; driven by *work* and *social insurance* topics

Constructing the Trade Shock: Prefecture-Level Tariff Changes

Key variable: Change in tariffs on Chinese exports imposed by importing countries

Industry-level tariff: Weighted average of destination-country applied tariffs (World Bank TRAINS), weights = 1995 export shares:

$$\tau_{jt}^X = \sum_n \frac{X_{j,1995}^{cn}}{\sum_{n'} X_{j,1995}^{cn'}} \cdot \tau_{jt}^{cn}$$

Prefecture-level tariff change (Kovak, 2013 style Bartik measure):

$$\Delta \text{tariffs on exports}_{it} = \sum_j \beta_{ij} [\ln(1 + \tau_{jt}^X) - \ln(1 + \tau_{jt}^X)]$$

$$\text{where } \beta_{ij} = \frac{\lambda_{ij}/\theta_{ij}}{\sum_{j'} \lambda_{ij'}/\theta_{ij'}}$$

- ▶ $\lambda_{ij} = L_{ij}/L_i$: employment share of industry j in prefecture i
- ▶ $1 - \theta_{ij}$: cost share of labor in industry j (from 2000 Industrial Enterprises Survey)
- ▶ Weights β_{ij} adjust for labor intensity, following the specific-factors logic of Kovak (2013)

Additional Export Demand Shocks

Two other WTO-induced positive demand shocks:

1. PNTR Shock (Pierce & Schott, 2016; Handley & Limão, 2017)

- ▶ Pre-WTO: U.S. applied MFN tariffs but status reviewed annually → **policy uncertainty**
- ▶ WTO made MFN status permanent
- ▶ Measure:

$$\text{PNTR}_{it} = \sum_j \beta_{ij} (\text{Col2}_j - \text{MFN}_j) \times \frac{\text{exp}_i^{\text{US}}}{\text{exp}_i^{\text{W}}}$$

2. MFA Quota Removal (Khandelwal et al., 2013)

- ▶ Textile/clothing exports to US, EU, Canada under quota until Jan. 2005
- ▶ Removal → boosted Chinese exports in these industries
- ▶ Measure:

$$\text{MFA}_{it} = \sum_p \frac{\text{exp}_{p,i,2000}}{\sum_{p'} \text{exp}_{p',i,2000}} \cdot \mathbf{1}(\text{MFA}_p)$$

Why focus on export tariffs?

- ▶ Decline was *direct and salient* from officials' perspective
- ▶ China's comparative advantage in labor-intensive goods → export expansion triggers labor demand → migration regulation change
- ▶ Import competition mostly on high-end goods (autos, agriculture)
- ▶ Empirically: controls for Δ tariffs on imports and intermediate inputs included

Identification Strategy: Long-Difference DiD

Econometric specification (long-difference, 2001–2007):

$$\Delta \sinh^{-1}(\text{regulation score})_{it} = \alpha_0 + \alpha_1 \underbrace{\Delta \text{tariffs on exports}_{it}}_{\text{Bartik measure}} + X_{it}\Gamma + \epsilon_{it}$$

- ▶ $\sinh^{-1}(\cdot)$: inverse hyperbolic sine transformation (handles zeros/negatives; score distribution very skewed)
- ▶ X_{it} : changes in tariffs on imports/inputs, initial regulation score, PNTR shock, MFA shock, lagged GDP/wage growth
- ▶ Standard errors: clustered at province level (+ bootstrap, AKM corrections)

Key identification assumption:

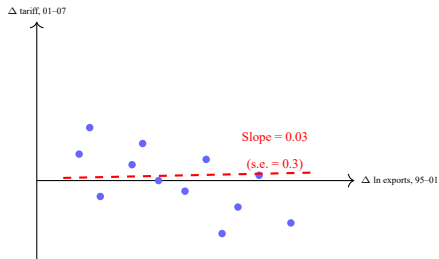
Conditional on X_{it} , no unobservables correlated with $\Delta \text{tariffs on exports}_{it}$ that *directly* affect regulation changes.

Why is this credible?

- ▶ Post-WTO tariff cuts determined by WTO rules and importing countries' schedules → **exogenous to Chinese prefecture conditions**

Identification: Pre-Trend Tests

Industry-Level Exogeneity

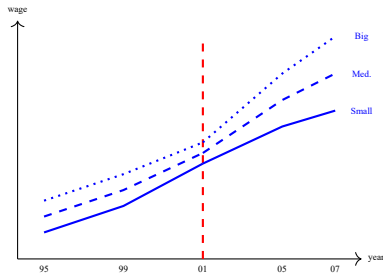


Industries with bigger pre-WTO export growth did **not** get bigger post-WTO tariff cuts

Industry-Level:

- ▶ Post-WTO tariff $\perp$ pre-WTO export growth
- ▶ Post-WTO tariff $\perp$ pre-WTO tariff decline

Prefecture-Level Pre-Trends



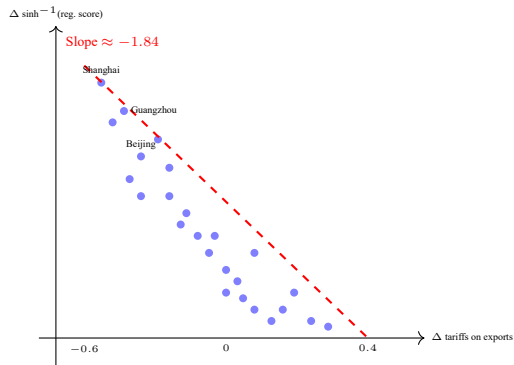
Wages parallel pre-WTO across small/medium/big tariff-decline groups. Divergence only *after* 2001.

Prefecture-Level:

- ▶ Wage trends: parallel pre-WTO
- ▶ GDP trends: slight divergence (controlled in regressions)

Main Result: Trade Liberalization → Pro-Migrant Regulations

Figure 5: Scatter plot — Δ Tariffs vs. Δ Regulation Score, 2001–2007



Key finding:

Negative and significant relationship: prefectures with larger tariff declines → larger regulation score increases

Magnitudes (Table 1, Col. 1):

- ▶ $\hat{\alpha}_1 = -1.04^{***}$ (s.e. = 0.30)
- ▶ 1 p.p. larger tariff decline → 1.04 larger $\Delta \sinh^{-1}(\text{score}) \approx 1$ s.d.

Economic magnitude:

- ▶ Big-decline prefectures: avg. -0.36 p.p. tariff cut
- ▶ Small-decline: avg. +0.04 p.p.
- ▶ Difference: 0.40 p.p. → **0.40 s.d.** larger regulation score increase

Table 1: Main Regression Results (Selected Columns)

	(1)	(2)	(5)	(6)	(7)
<i>Panel A. Dep. var.: $\Delta \sinh^{-1}(\text{regulation score})$, manual coding</i>					
Δ tariffs on exports, 01–07	−1.04*** (0.30)	−0.91*** (0.26)	−1.60*** (0.50)	−1.38*** (0.49)	−1.02* (0.53)
PNTR shock		4.67*** (1.44)		2.89** (1.27)	1.03 (1.21)
MFA shock		5.43*** (1.94)		4.20** (1.91)	4.86** (2.25)
Δ log wage, 95–01					0.69
Δ log GDP p.c., 95–01					0.65***
Prefectures	333	333	248	248	235
R^2	0.09	0.14	0.08	0.11	0.15
Bootstrap p	.006	.005	.006	.014	.078
AKM(0) p	.024	.020	.036	.055	.051
<i>Panel B. Random Forest coding: $\hat{\alpha}_1$ ranges from −1.39*** to −0.89*</i>					

All columns control for Δ tariffs on imports, Δ tariffs on inputs, initial score. SE clustered at province.
Cols 5–7 restrict to 248 prefectures in Prefecture Statistics Yearbook (more urban).

Heterogeneous Effects: Who Responds More?

Specification:

$$\Delta \sinh^{-1}(\text{reg. score})_{it} = \beta_0 + \beta_1 \Delta T_{it} + \beta_2 I_{it} + \beta_3 \underbrace{I_{it} \times \Delta T_{it}}_{\text{interaction}} + X_{it} \Gamma + \epsilon_{it}$$

Baseline characteristic I :	Migrant intensity	Private firm share	GDP p.c.	Wage
$\hat{\beta}_1 (\Delta T)$	4.65 (3.65)	1.21 (0.72)	10.98* (5.97)	29.53 (18.72)
$\hat{\beta}_3 (\Delta T \times I)$	-17.31 (10.93)	-5.47*** (1.09)	-1.29* (0.68)	-3.33 (2.07)
Effect at mean I : N	-1.24 248	-1.80 248	-0.55 248	-0.81 248

Interpretation:

- ✓ **Migrant-intensive** prefectures respond more strongly
- ✓ Higher **private-firm share** → stronger response (SOEs hire fewer migrants)
- ✓ **Richer** prefectures respond more (higher income $\approx$ higher migrant demand)

Reinforces the mechanism: prefectures that *needed* migrants most were the ones that responded to tariff cuts by relaxing Hukou restrictions

Do Regulations Matter? Evidence on Migrant Well-Being (2005)

Question: Are these *de jure* regulation changes also *de facto* effective?

Data: 2005 1% Population Survey (cross-section)

Dep. var. (migrants):	Unemp. Ins. rate	Pension rate	Medical Ins. rate	Contract (months)	Monthly Income	School Enroll.
$\sinh^{-1}(\text{reg. score})_{2005}$	0.01* (0.01)	0.02** (0.01)	0.02** (0.01)	0.09 (0.10)	29.27** (13.98)	0.00 (0.00)
Local well-being control	Yes	Yes	Yes	Yes	Yes	Yes
Mean (migrants)	0.21	0.34	0.36	4.57	923	0.94
Mean (locals)	0.37	0.60	0.59	5.05	981	0.95

Key takeaways:

- ▶ 1 s.d. $\uparrow$ regulation score: unemployment insurance gap closes by **6.25%**; income rises **29 yuan/month** ($\approx 50\%$ of migrant-local wage gap)
- ▶ Effects significant for *workplace* outcomes (insurance, wages); *not* for school enrollment
- ▶ Local well-being *not* significantly affected (Cols 1–3, 5 in Panel B) $\rightarrow$ regulation captures *migrant-specific* improvement

From Regulations to Migrant Flows

Reduced-form: $\Delta Y_{it} = \gamma_0 + \gamma_1 \Delta \text{tariffs on exports}_{it} + X_{it}\Gamma + \xi_{it}$

Association: $\Delta Y_{it} = \pi_0 + \pi_1 \Delta \sinh^{-1}(\text{reg. score})_{it} + X_{it}\Phi + \zeta_{it}$

	Δ Migrant share of pop.			$\Delta \ln \#$ migrants (short dist.)		
	(1)	(2)	(3)	(4)	(5)	(6)
Δ tariffs on exports	-7.11** (3.39)		-6.23* (3.22)	-1.34*** (0.31)		-1.18*** (0.27)
$\Delta \sinh^{-1}(\text{reg. score})$		1.03*** (0.29)	0.87*** (0.22)		0.18*** (0.04)	0.20*** (0.04)

Results by Migration Distance:

- ▶ **Short-distance:** Both trade and regulations significant
- ▶ **Medium-distance:** Trade effect significant; regulation effect marginally significant
- ▶ **Long-distance:** Trade effect *insignificant*; regulation effect *significant*

⇒ For long-distance migration, **regulatory amenities may matter more than prices**

Caveat: π_1 is not causal (simultaneous determination, reverse causality). See Appendix C.11 for IV attempts using natural pop. growth rate.

Robustness Checks: A Comprehensive Battery

Measurement Robustness:

- ✓ NLP codings (RF, MLP, CNN, average): Table C1 — nearly identical results
- ✓ 3-point scale ($\{-1, 0, 1\}$) instead of 5-point: Table C4
- ✓ Score levels instead of $\sinh^{-1}$: Table C3
- ✓ Topic decomposition: work & public goods most affected (Table C4, Cols 4–8)

Specification Robustness:

- ✓ Stacked long-difference (1995–01 + 2001–07 panel): Table C2
- ✓ Alternative Bartik weights (λ_{ij} only): Table C5, Col 2
- ✓ Autor et al. (2013) style IPW measures: Table C6, Col 1

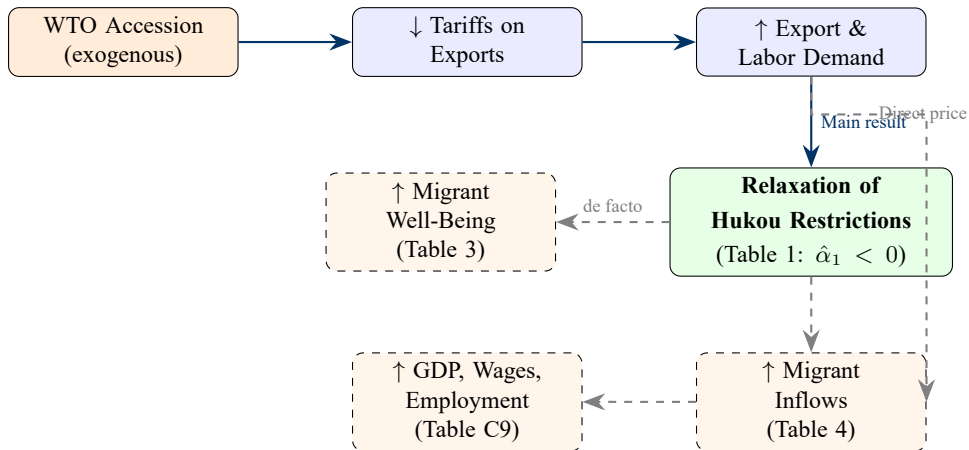
Identification Robustness:

- ✓ Adding industry employment shares one-at-a-time: Figure C4 — stable around -1.60
- ✓ Bootstrap SEs and AKM(0), AKM(1) corrections: similar p -values
- ✓ Pre-WTO placebo: slopes ≈ 0 , insignificant (Figure C2)

Strategic Interaction:

- ✓ Prefectures respond to tariff & regulation changes in *same-province* peers (Table C6)
- ✓ Strongest competition among prefectures with similar *GDP levels* (Table C7)
- ▶ Consistent with Tiebout-style fiscal competition within income tiers

The Economic Logic: Putting It All Together



Central insight: Trade liberalization affects the economy not only through *prices* but also through *institutional change*. Labor market frictions are **endogenous** to trade policy.

Conclusion and Open Questions

Summary of Findings:

- 1 WTO-induced tariff reductions on Chinese exports caused local governments to enact **pro-migrant regulations** (0.40 s.d. difference between high- and low-exposure prefectures)
- 2 Effect is stronger for migrant-intensive, private-sector-dominated, richer prefectures
- 3 Regulations associated with improved migrant well-being (insurance, wages) and increased migrant inflows
- 4 Prefectures *compete* for migrants with similar-income peers

Open Questions and Limitations:

- ▶ Cannot cleanly separate direct price channel from indirect regulation channel on migrant flows
- ▶ De jure $\neq$ de facto: only cross-sectional evidence on enforcement (2005)
- ▶ Welfare implications: Is inter-prefecture competition efficiency-enhancing or a race to the bottom?
- ▶ Broader institutional effects of WTO: rule of law, transparency, property rights?
- ▶ External validity: Does the mechanism extend to other countries with internal migration barriers (Vietnam, Russia)?

Thank You

Questions & Discussion

Paper: Tian, Y. (2024). “International Trade Liberalization and Domestic Institutional Reform.”
The Review of Economics and Statistics, 106(3): 794–813.

DOI: 10.1162/rest_a_01175